Recursion: The Mirrors

Chapter 2

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Recursive Solutions

- A recursive function calls itself
- Each recursive call solves an identical, but smaller, problem
- Test for base case enables recursive calls to stop
- Eventually, one of smaller problems must be the base case
- Some recursive solutions are impractical because they are so inefficient
- Can provide elegantly simple solutions to problems of great complexity

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Recursive Solutions

Questions for constructing recursive solutions

1. How to define the problem in terms of a smaller problem of same type?
2. How does each recursive call diminish the size of the problem?
3. What instance of problem can serve as base case?
4. As problem size diminishes, will you reach base case?

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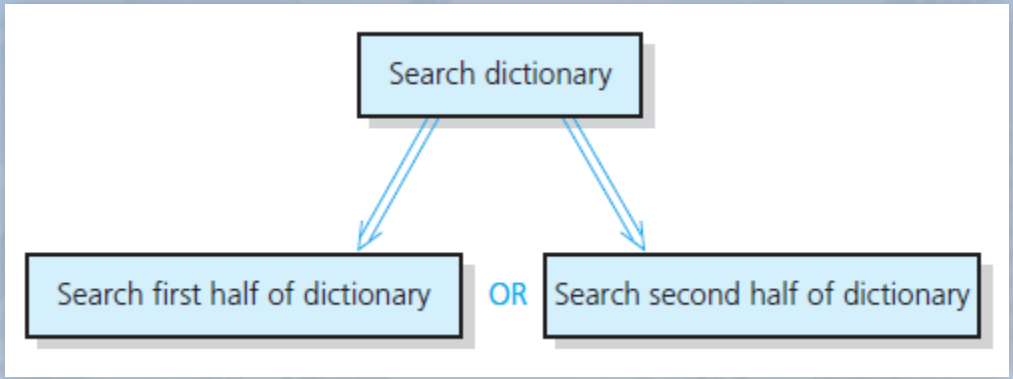
Recursive Solutions

- Recursion breaks problem into smaller identical problems
  - An alternative to iteration
- FIGURE 2-1 A recursive solution



- This strategy is called divide and conquer.
- Solve the dictionary search problem by first dividing the dictionary into two halves and then conquering the appropriate half.
- Solve the smaller problem by using the same divide-and-conquer strategy.
- Dividing continues until you reach the base case.

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A Recursive Valued Function: The Factorial of *n*

- An iterative solution

$$factorial(n) = n \times (n-1) \times (n-2) \times \cdots \times 1 \quad \text{for an integer } n > 0$$
$$factorial(0) = 1$$

- A recursive solution

$$factorial(n) = \begin{cases} 1 & if\ n = 0 \\ n \times factorial(n-1) & if\ n > 0 \end{cases}$$

Note: Do not use recursion if a problem has a simple, efficient iterative solution

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A Recursive Valued Function: The Factorial of *n*

$$factorial(n) = \begin{cases} 1 & if\ n = 0 \\ n \times factorial(n-1) & if\ n > 0 \end{cases}$$

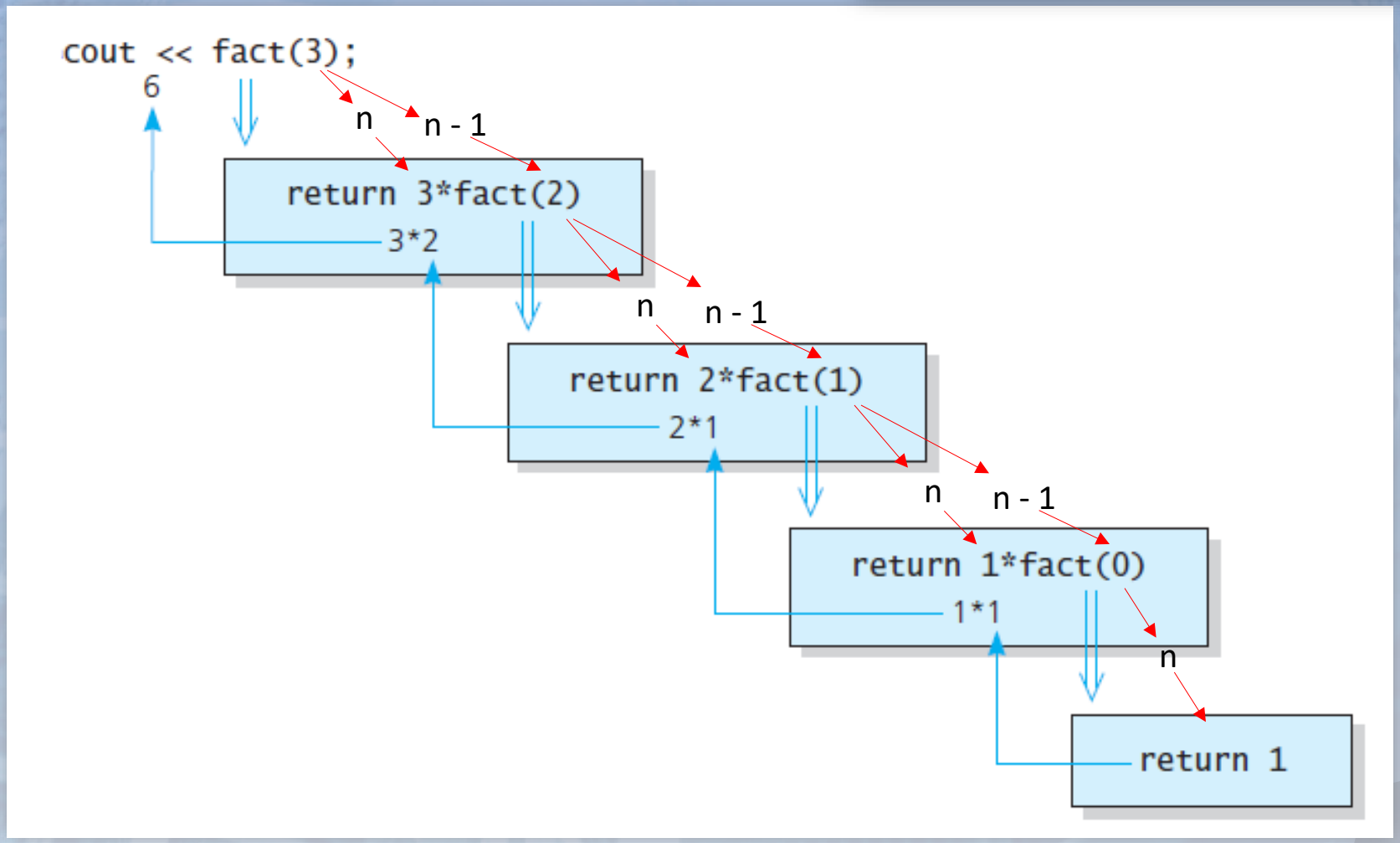

FIGURE 2-2 fact(3)

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The Box Trace

1. Label each recursive call
2. Represent each call to function by a new box
3. Draw arrow from box that makes call to newly created box
4. After you create new box, execute body of function
5. On exiting function, cross off current box and follow its arrow back

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The Box Trace

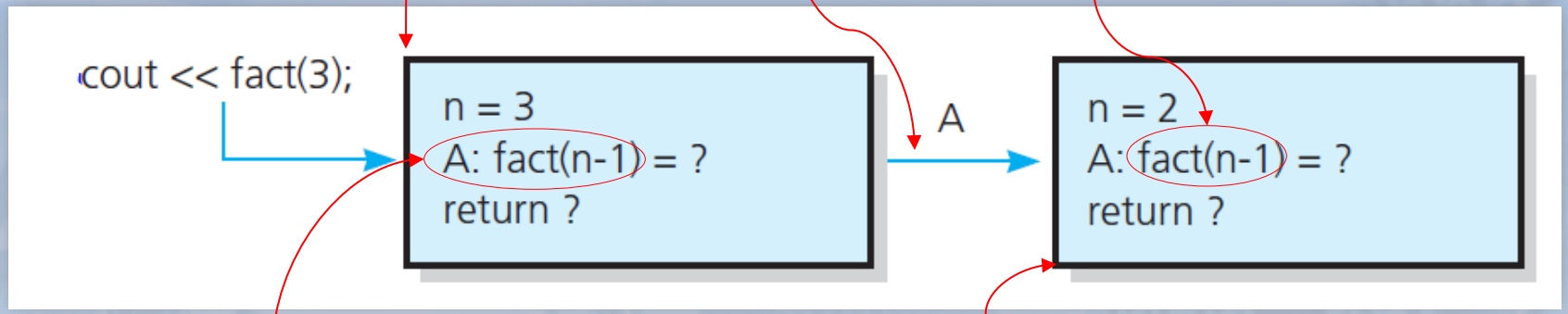
```
int fact(int n) {
  if (n==0) {
    return 1;}
  else {
    return n * fact(n-1);}}
```

Each box is activation record – typical compiler implementation of a function call

5. On exiting the function, cross off the current box and follow its arrow back to the box that called the function.

3. Draw arrow to newly created box

4. Execute function



1. Label each recursive call - you return to point A after each recursive call, substitute the computed value, and continue evaluating n*fact(n-1)

2. Represent each call to function by a new box – update local environment (arg values, local variables, placeholder for value returned (label placeholder to correspond to step 1 label, and value of function itself

FIGURE 2-4 The beginning of the box trace

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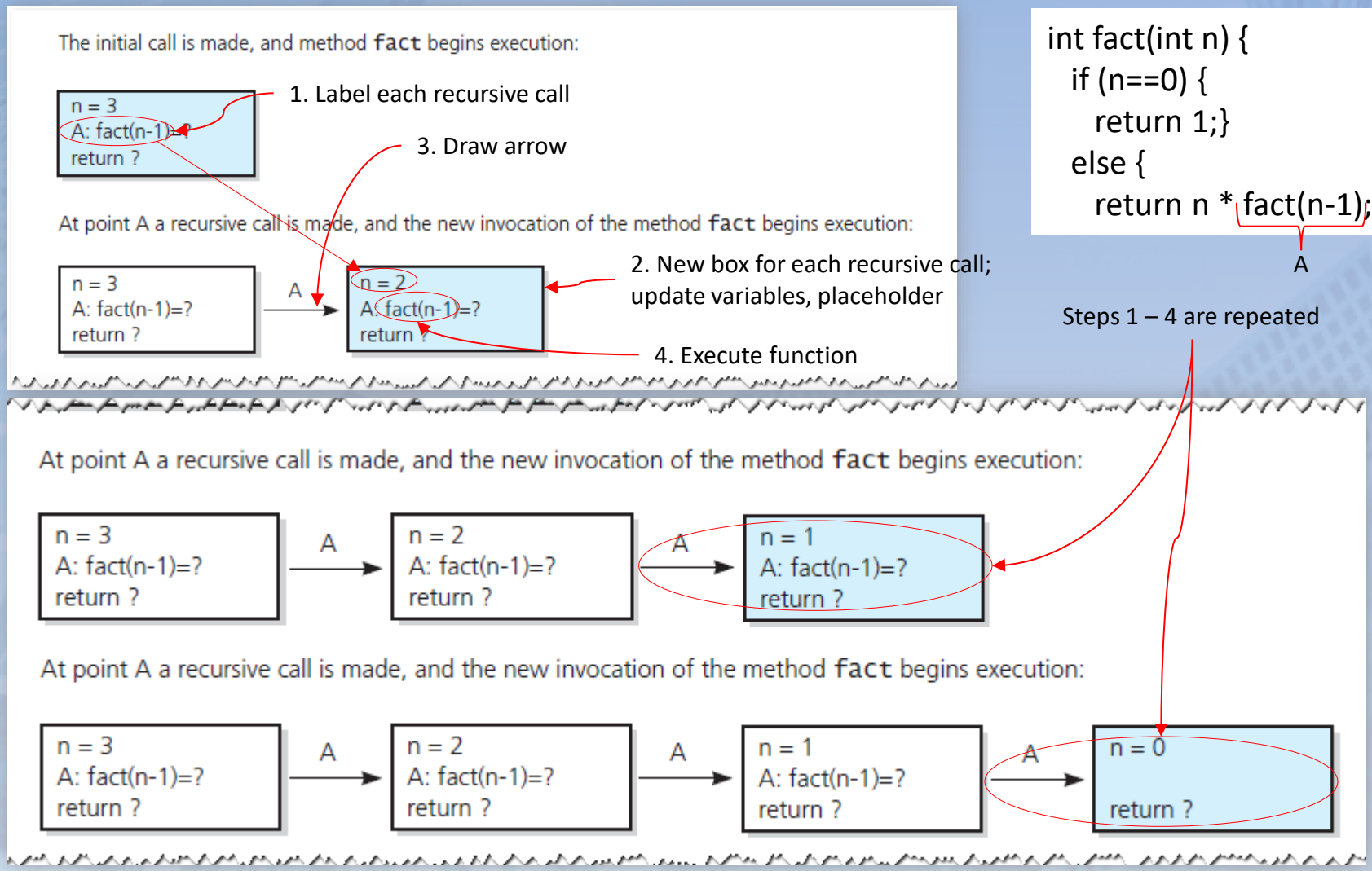
The Box Trace




```
int fact(int n) {
  if (n==0) {
    return 1;}
  else {
    return n * fact(n-1);}}
```

A

Steps 1 – 4 are repeated

FIGURE 2-5 Box trace of fact(3)

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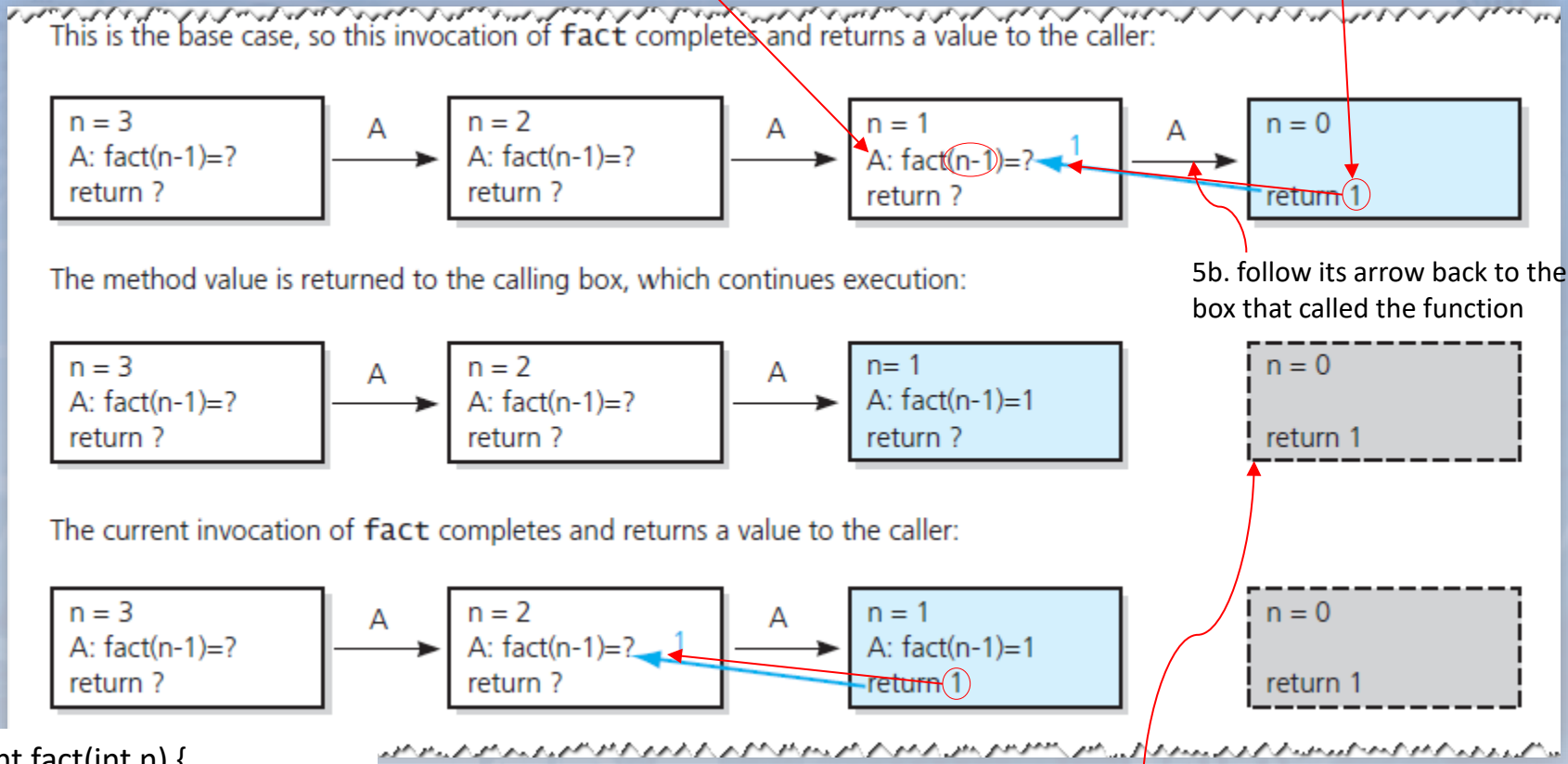
The Box Trace

5c. This box now becomes the current box, and the label on the arrow specifies the exact location at which execution of the function should continue.

5d. Substitute the value returned by the just-terminated function call for the appropriate placeholder in the current box.



5b. follow its arrow back to the box that called the function

5a. On exiting the function, cross off the current box

```
int fact(int n) {
  if (n==0) {
    return 1;}
  else {
    return n * fact(n-1);}}
```

A

FIGURE 2-5 Box trace of fact(3)

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The Box Trace

```
int fact(int n) {
  if (n==0) {
    return 1;}
  else {
    return n * fact(n-1);}}
```

A

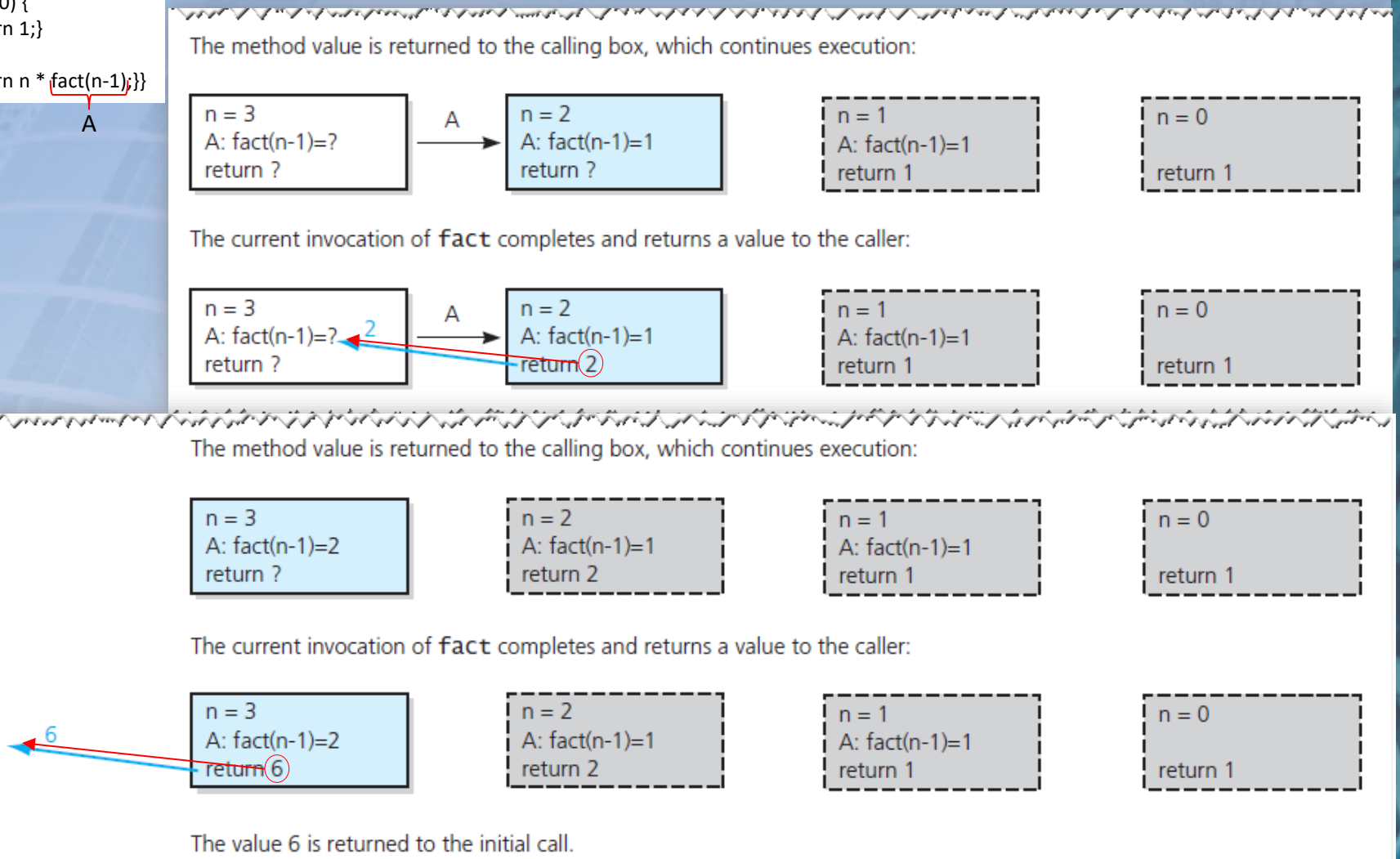

FIGURE 2-5 Box trace of fact(3)

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The Binary Search

Consider details before implementing algorithm:

1. How to pass half of anArray to recursive calls of binarySearch ?
2. How to determine which half of array contains target?
3. What should base case(s) be?
4. How will binarySearch indicate result of search?

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The Binary Search

```
binarySearch(anArray: ArrayType, target: ValueType)

    if (anArray is of size 1)
        Determine if anArray's value is equal to target
    else
    {
        Find the midpoint of anArray
        Determine which half of anArray contains target
        if (target is in the first half of anArray)
            binarySearch(first half of anArray, target)
        else
            binarySearch(second half of anArray, target)
    }
```

High level pseudocode for binarySearch with anArray of sorted integers

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The Binary Search

```
/** Searches the array anArray[first] through anArray[last]
    for a given value by using a binary search.
 @pre  0 <= first, last <= SIZE - 1, where SIZE is the
       maximum size of the array, and anArray[first] <=
       anArray[first + 1] <= ... <= anArray[last].
 @post  anArray is unchanged and either anArray[index] contains
        the given value or index == -1.
 @param anArray  The array to search.
 @param first  The low index to start searching from.
 @param last  The high index to stop searching at.
 @param target  The search key.
 @return  Either index, such that anArray[index] == target, or -1.
*/
int binarySearch(const int anArray[], int first, int last, int target)
{
   int index;
   if (first > last)
      index = -1;   // target not in original array
   else
   {
      // If target is in anArray,
      // anArray[first] <= target <= anArray[last]
      int mid = first + (last - first) / 2;
      if (target == anArray[mid])
         index = mid; // target found at anArray[mid]
      else if (target < anArray[mid])
         // Point X
         index = binarySearch(anArray, first, mid - 1, target);
      else
         // Point Y
         index = binarySearch(anArray, mid + 1, last, target);
   }  // end if

   return index;
}  // end binarySearch
```

Recursive binarySearch function with anArray of sorted integers

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The Binary Search

```
int binarySearch(const int anArray[], int first, int last, int target)
{
   int index;
   if (first > last)
      index = -1;   // target not in original array
   else
   {
      // If target is in anArray,
      // anArray[first] <= target <= anArray[last]
      int mid = first + (last - first) / 2;
      if (target == anArray[mid])
         index = mid; // target found at anArray[mid]
      else if (target < anArray[mid])
         // Point X
         index = binarySearch(anArray, first, mid - 1, target);
      else
         // Point Y
         index = binarySearch(anArray, mid + 1, last, target);
   }  // end if

   return index;
}  // end binarySearch
```

FIGURE 2-10 Box traces of binarySearch with **anArray = <1, 5, 9, 12, 15, 21, 29, 31>:** (a) a successful search for 9;

Return index of found item

Item we are looking for

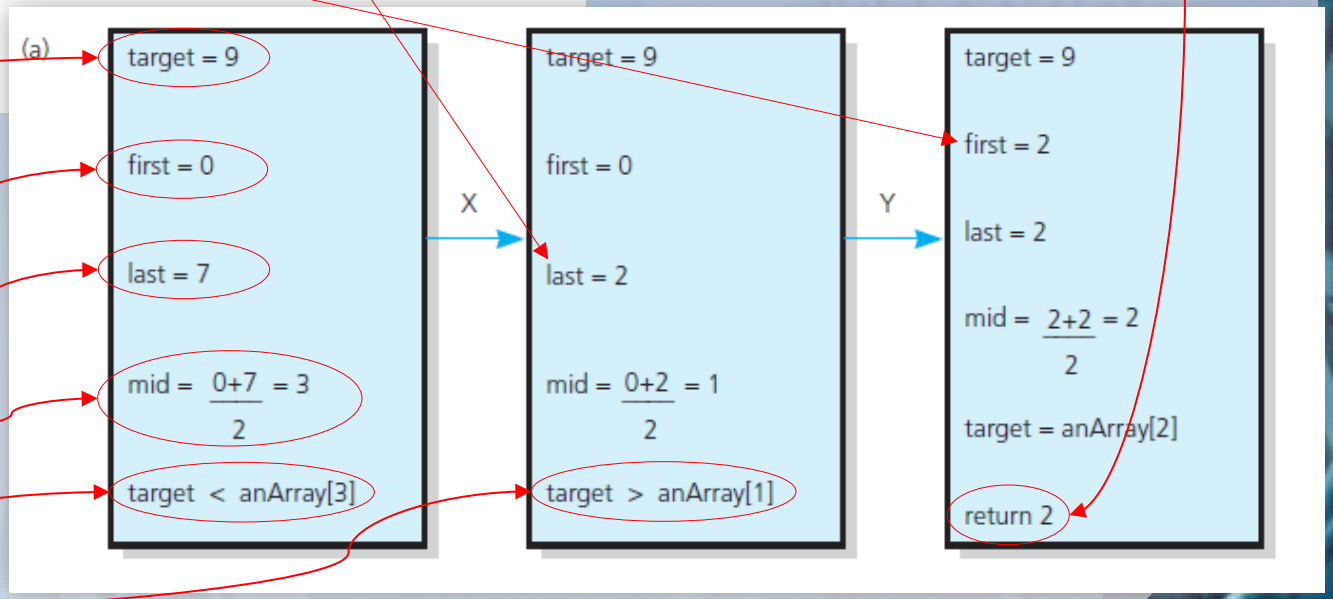
1st index of array

Last index of array

Middle index of array

true (9 < 12)

true (9 > 5)



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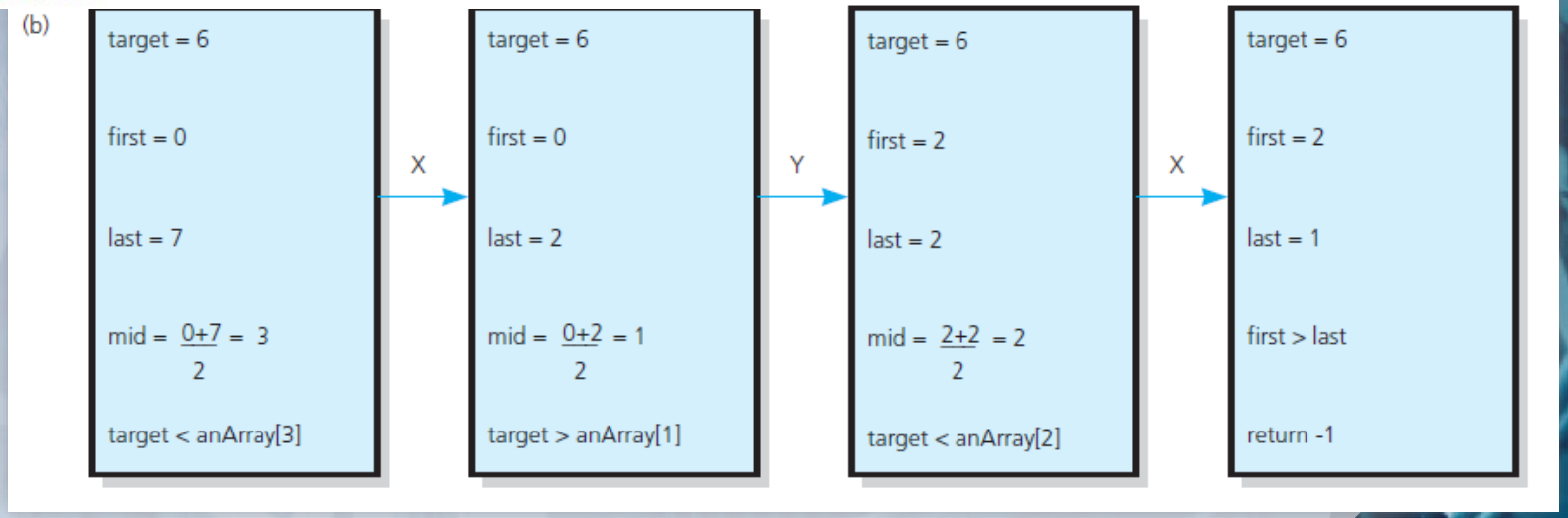
The Binary Search

```
int binarySearch(const int anArray[], int first, int last, int target)
{
   int index;
   if (first > last)
      index = -1;   // target not in original array
   else
   {
      // If target is in anArray,
      // anArray[first] <= target <= anArray[last]
      int mid = first + (last - first) / 2;
      if (target == anArray[mid])
         index = mid; // target found at anArray[mid]
      else if (target < anArray[mid])
         // Point X
         index = binarySearch(anArray, first, mid - 1, target);
      else
         // Point Y
         index = binarySearch(anArray, mid + 1, last, target);
   }  // end if

   return index;
}  // end binarySearch
```

FIGURE 2-10 Box traces of binarySearch with anArray = <1, 5, 9, 12, 15, 21, 29, 31>: (b) an unsuccessful search for 6



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Recursion and Efficiency

- Factors that contribute to inefficiency
  - Overhead associated with function calls
  - Some recursive algorithms inherently inefficient
- Keep in mind
  - Recursion can clarify complex solutions … but …
  - Clear, efficient iterative solution may be better

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End

Chapter 2

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Appendix

Chapter 2

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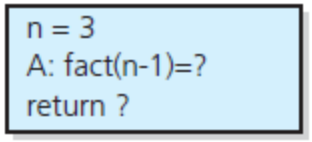
The Box Trace

The initial call is made, and method `fact` begins execution:



At point A a recursive call is made, and the new invocation of the method `fact` begins execution:

FIGURE 2-5 Box trace of fact(3)

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The Box Trace

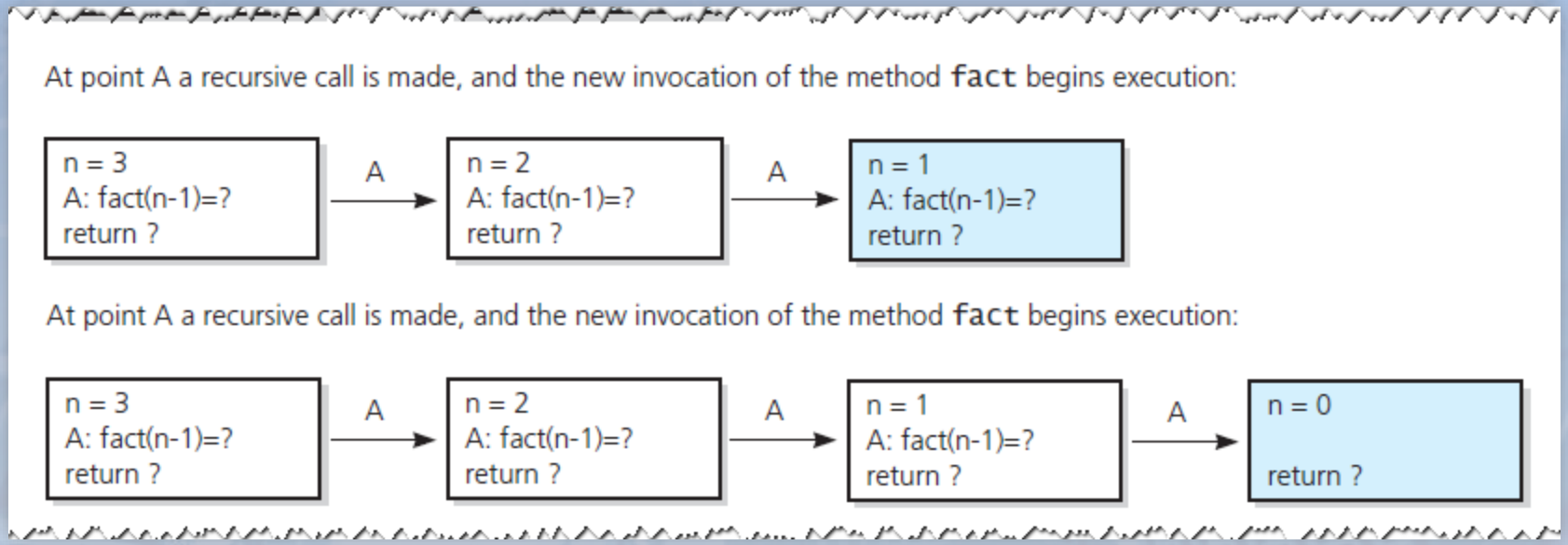

FIGURE 2-5 Box trace of fact(3)

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The Box Trace

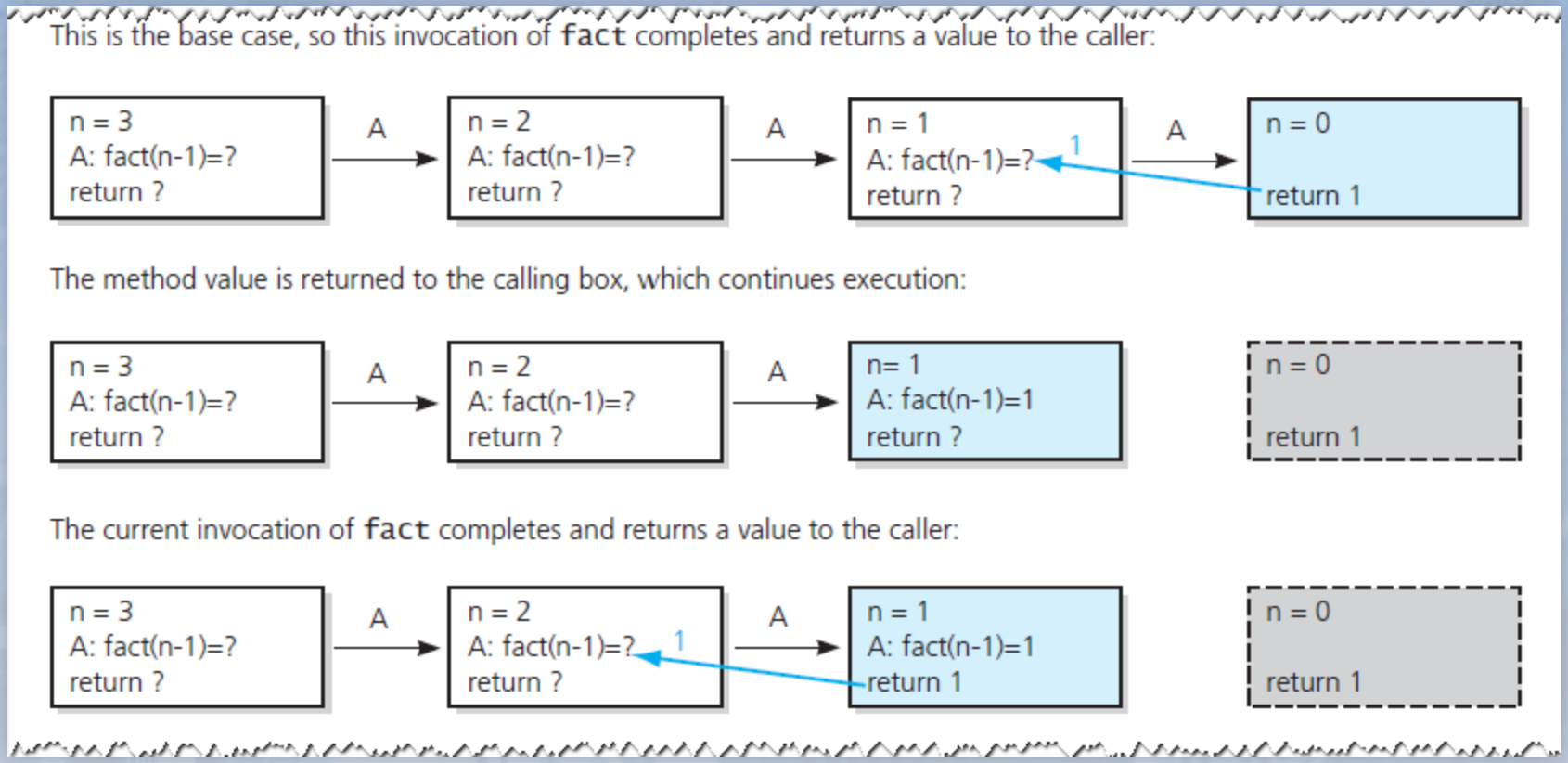

FIGURE 2-5 Box trace of fact(3)

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The Box Trace

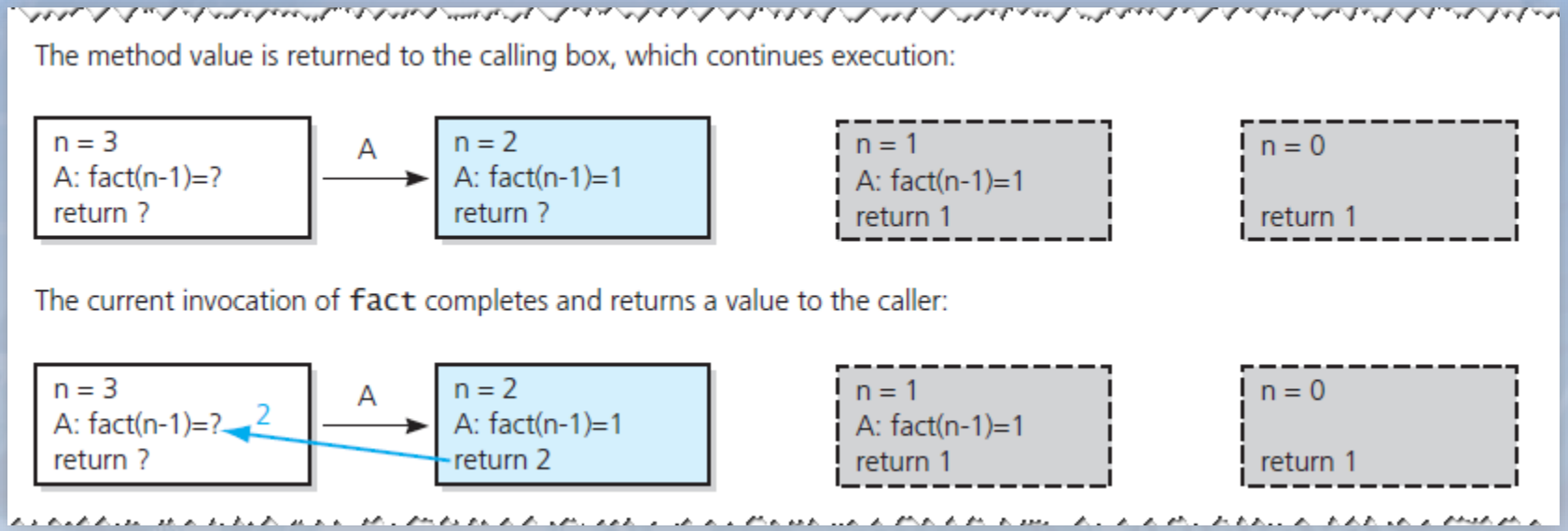

FIGURE 2-5 Box trace of fact(3)

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The Box Trace

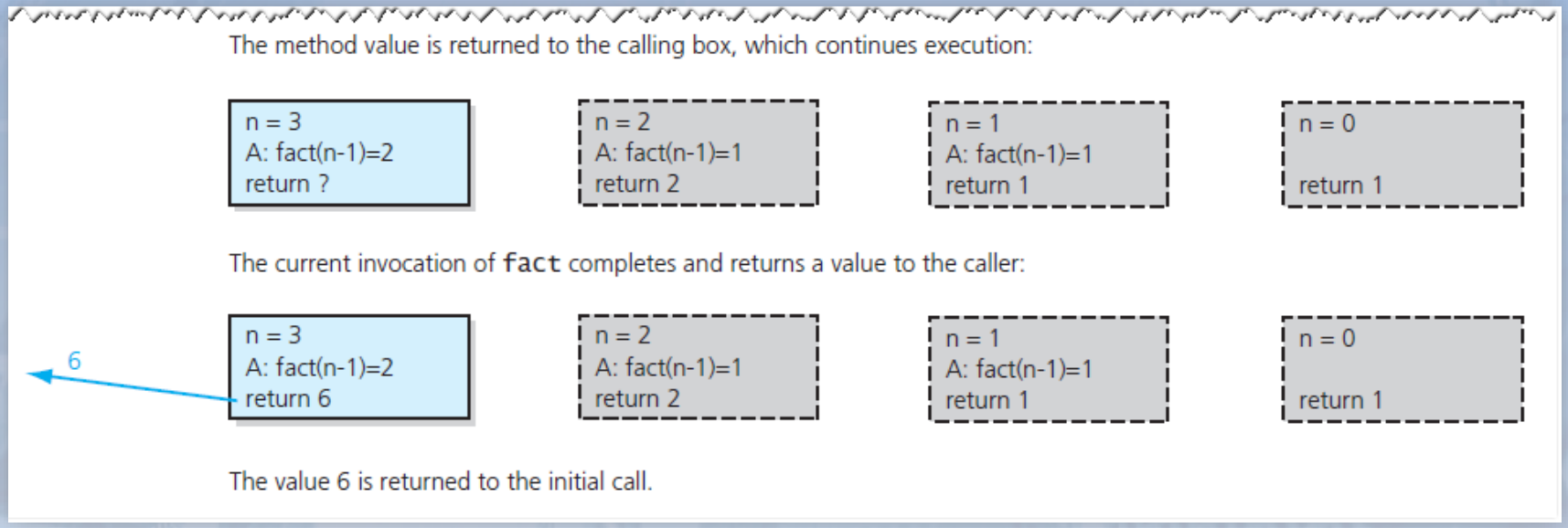

FIGURE 2-5 Box trace of fact(3)

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A Recursive Void Function: Writing a String Backward

- Likely candidate for minor task is writing a single character.
  - Possible solution: strip away the last character

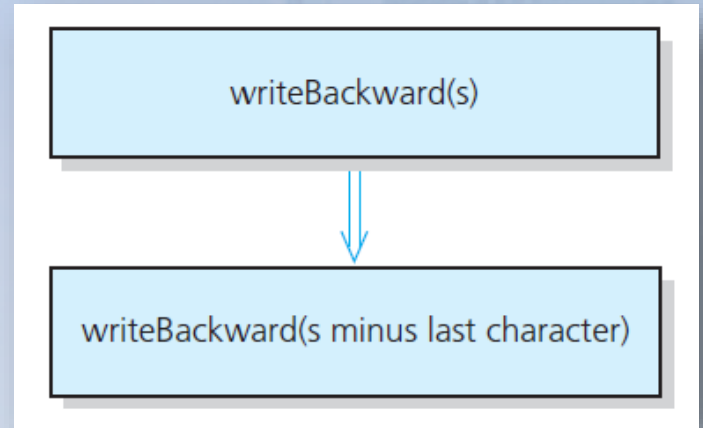

FIGURE 2-6 A recursive solution

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A Recursive Void Function: Writing a String Backward

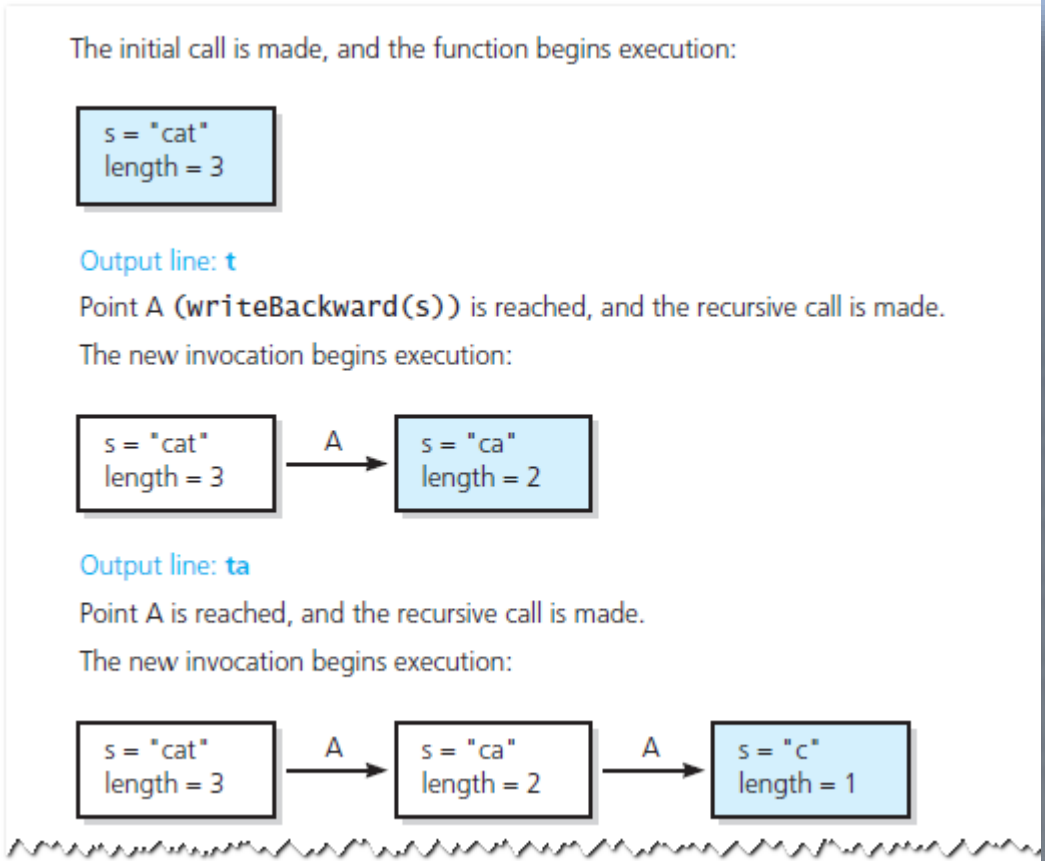

FIGURE 2-7 Box trace of writeBackward("cat")

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A Recursive Void Function: Writing a String Backward

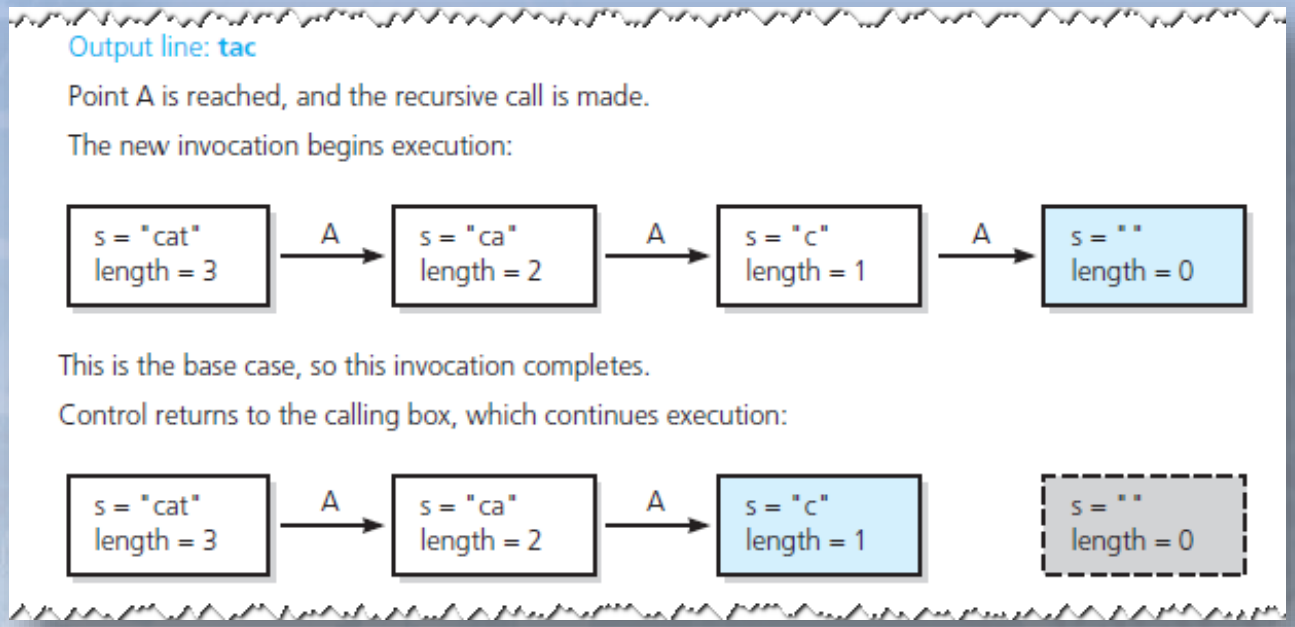

FIGURE 2-7 Box trace of writeBackward("cat")

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A Recursive Void Function: Writing a String Backward

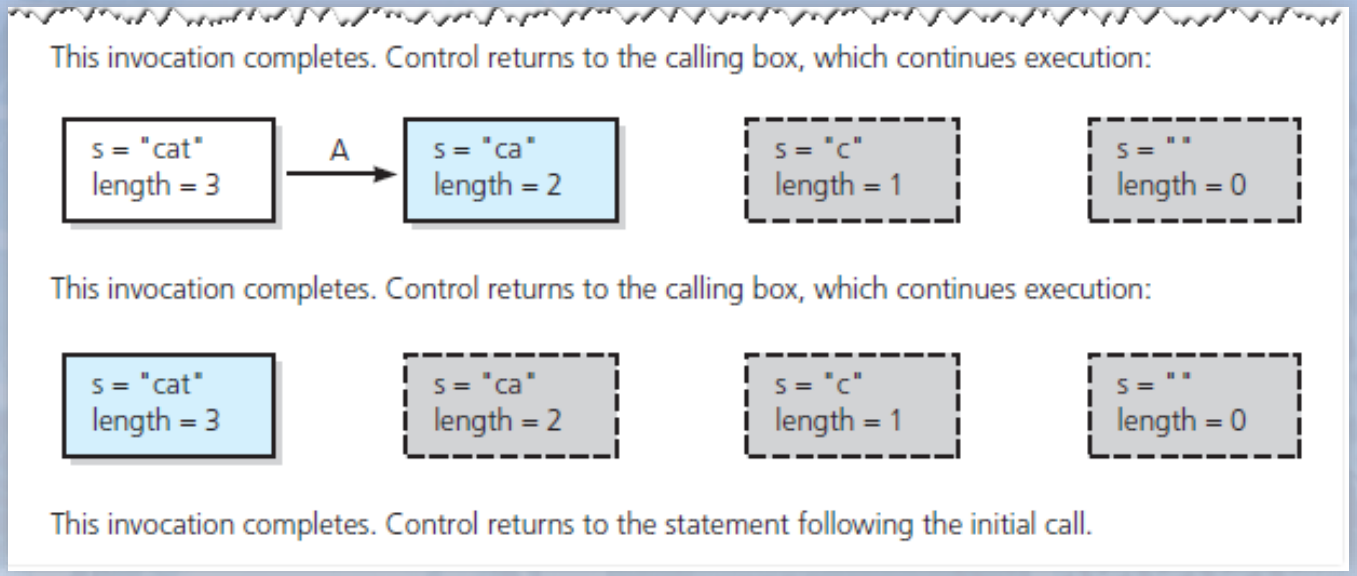

FIGURE 2-7 Box trace of writeBackward("cat")

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A Recursive Void Function: Writing a String Backward

- Another possible solution
  - Strip away the first character

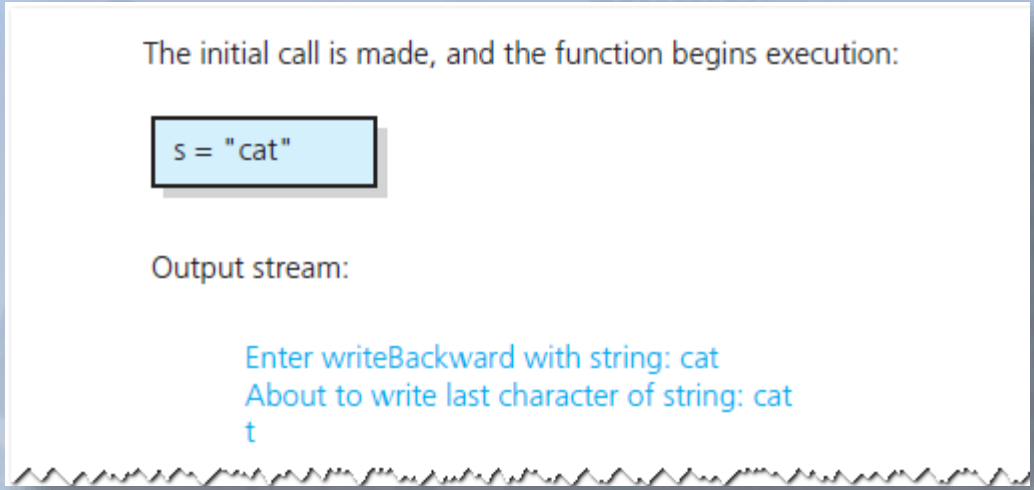

FIGURE 2-8 Box trace of writeBackward("cat") in pseudocode

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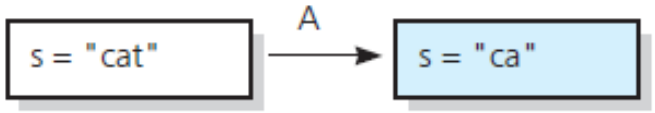
A Recursive Void Function: Writing a String Backward

Point A is reached, and the recursive call is made. The new invocation begins execution:



Output stream:

```
Enter writeBackward with string: cat
About to write last character of string: cat
t
Enter writeBackward with string: ca
About to write last character of string: ca
a
```

FIGURE 2-8 Box trace of writeBackward("cat") in pseudocode

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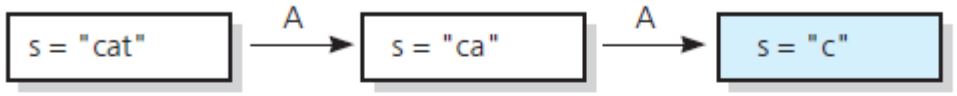
A Recursive Void Function: Writing a String Backward

Point A is reached, and the recursive call is made. The new invocation begins execution:



Output stream:

```
Enter writeBackward with string: cat
About to write last character of string: cat
t
Enter writeBackward with string: ca
About to write last character of string: ca
a
Enter writeBackward with string: c
About to write last character of string: c
c
```

FIGURE 2-8 Box trace of writeBackward("cat") in pseudocode

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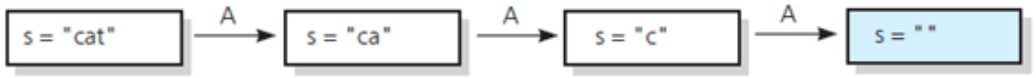
A Recursive Void Function: Writing a String Backward

Point A is reached, and the recursive call is made. The new invocation begins execution:



This invocation completes execution, and a return is made.

Output stream:

```
Enter writeBackward with string: cat
About to write last character of string: cat
t
Enter writeBackward with string: ca
About to write last character of string: ca
a
Enter writeBackward with string: c
About to write last character of string: c
c
Enter writeBackward with string:
Leave writeBackward with string:
```

FIGURE 2-8 Box trace of writeBackward("cat") in pseudocode

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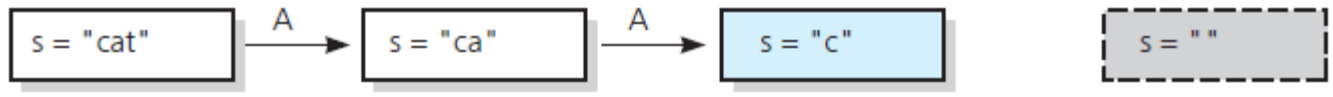
A Recursive Void Function: Writing a String Backward



This invocation completes execution, and a return is made.

Output stream:

```
Enter writeBackward with string: cat
About to write last character of string: cat
t
Enter writeBackward with string: ca
About to write last character of string: ca
a
Enter writeBackward with string: c
About to write last character of string: c
c
Enter writeBackward with string:
Leave writeBackward with string:
Leave writeBackward with string: c
```

FIGURE 2-8 Box trace of writeBackward("cat") in pseudocode

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A Recursive Void Function: Writing a String Backward

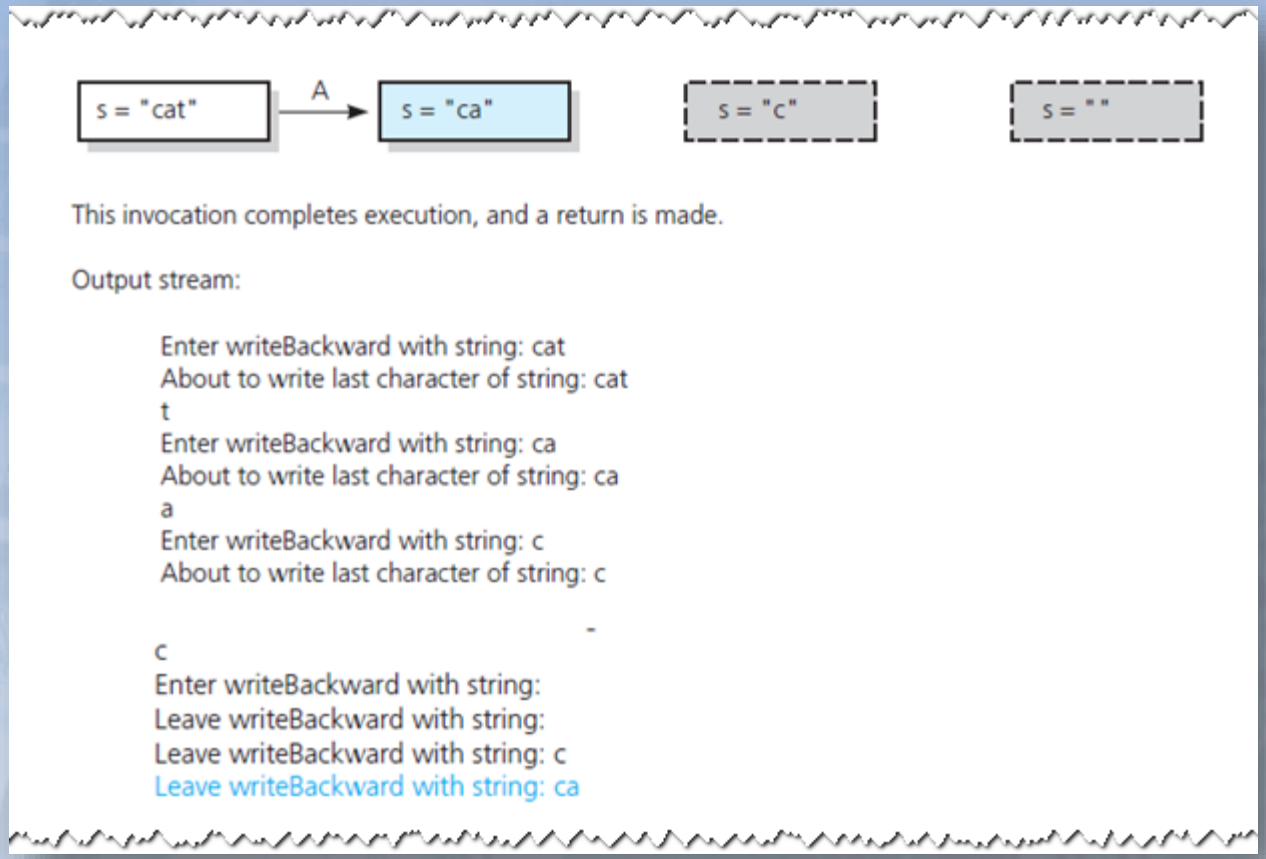

FIGURE 2-8 Box trace of writeBackward("cat") in pseudocode

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A Recursive Void Function: Writing a String Backward

This invocation completes execution, and a return is made.

Output stream:

```
Enter writeBackward with string: cat
About to write last character of string: cat
t
Enter writeBackward with string: ca
About to write last character of string: ca
a
Enter writeBackward with string: c
About to write last character of string: c
c
Enter writeBackward with string:
Leave writeBackward with string:
Leave writeBackward with string: c
Leave writeBackward with string: ca
```

FIGURE 2-8 Box trace of writeBackward("cat") in pseudocode

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A Recursive Void Function: Writing a String Backward

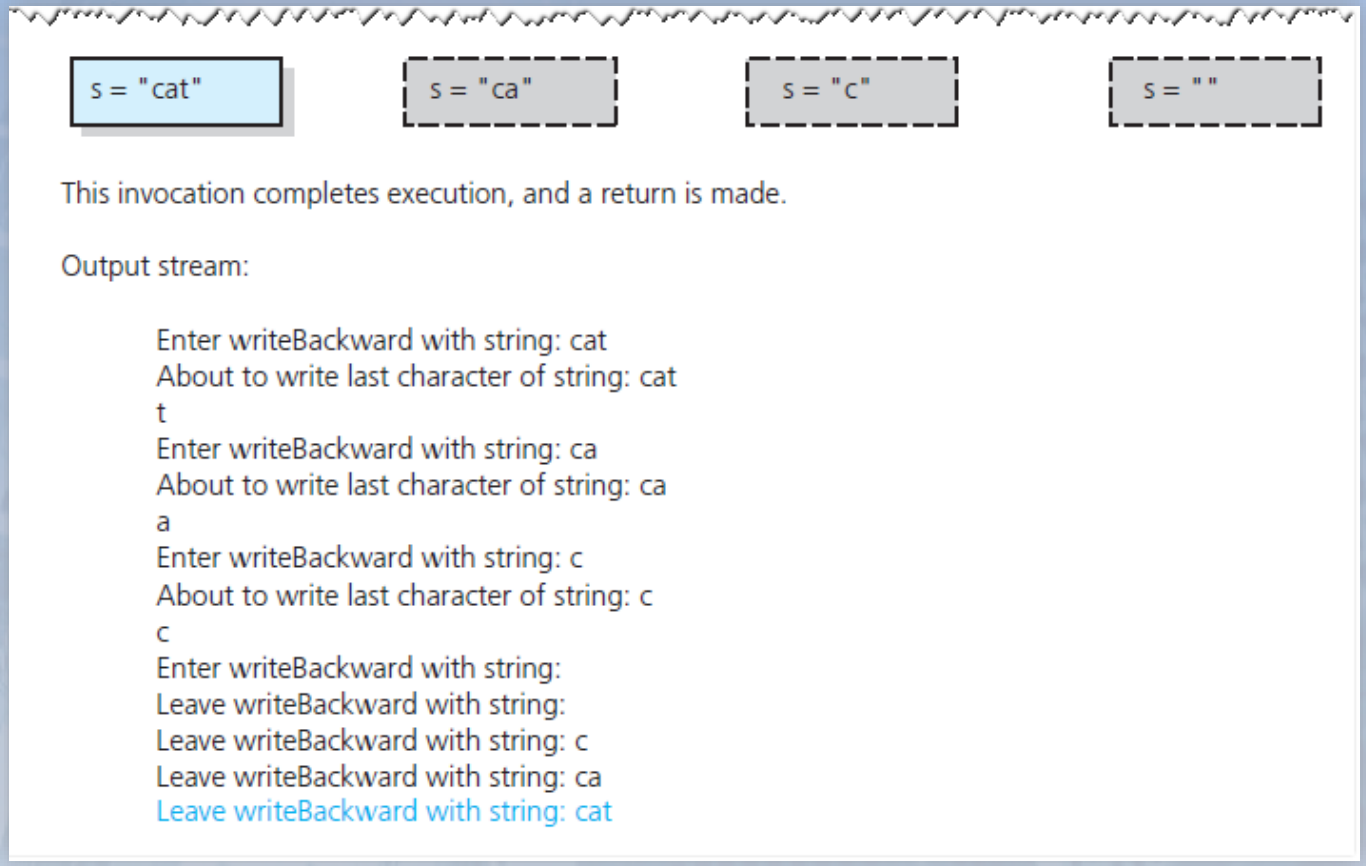

FIGURE 2-8 Box trace of writeBackward("cat") in pseudocode

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A Recursive Void Function: Writing a String Backward

The initial call is made, and the function begins execution:

s = "cat"

Output stream:

Enter writeBackward2 with string: cat

Point A is reached, and the recursive call is made. The new invocation begins execution:

s = "cat" —A→ s = "at"

Output stream:

Enter writeBackward2 with string: cat
Enter writeBackward2 with string: at

FIGURE 2-8 Box trace of writeBackward2("cat") in pseudocode

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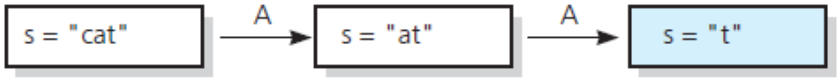
A Recursive Void Function: Writing a String Backward

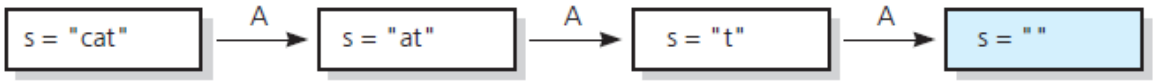
Point A is reached, and the recursive call is made. The new invocation begins execution:



Output stream:

```
Enter writeBackward2 with string: cat
Enter writeBackward2 with string: at
Enter writeBackward2 with string: t
```

Point A is reached, and the recursive call is made. The new invocation begins execution:



This invocation completes execution, and a return is made.

Output stream:

```
Enter writeBackward2 with string: cat
Enter writeBackward2 with string: at
Enter writeBackward2 with string: t
Enter writeBackward2 with string:
Leave writeBackward2 with string:
```

FIGURE 2-8 Box trace of writeBackward2("cat") in pseudocode

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A Recursive Void Function: Writing a String Backward

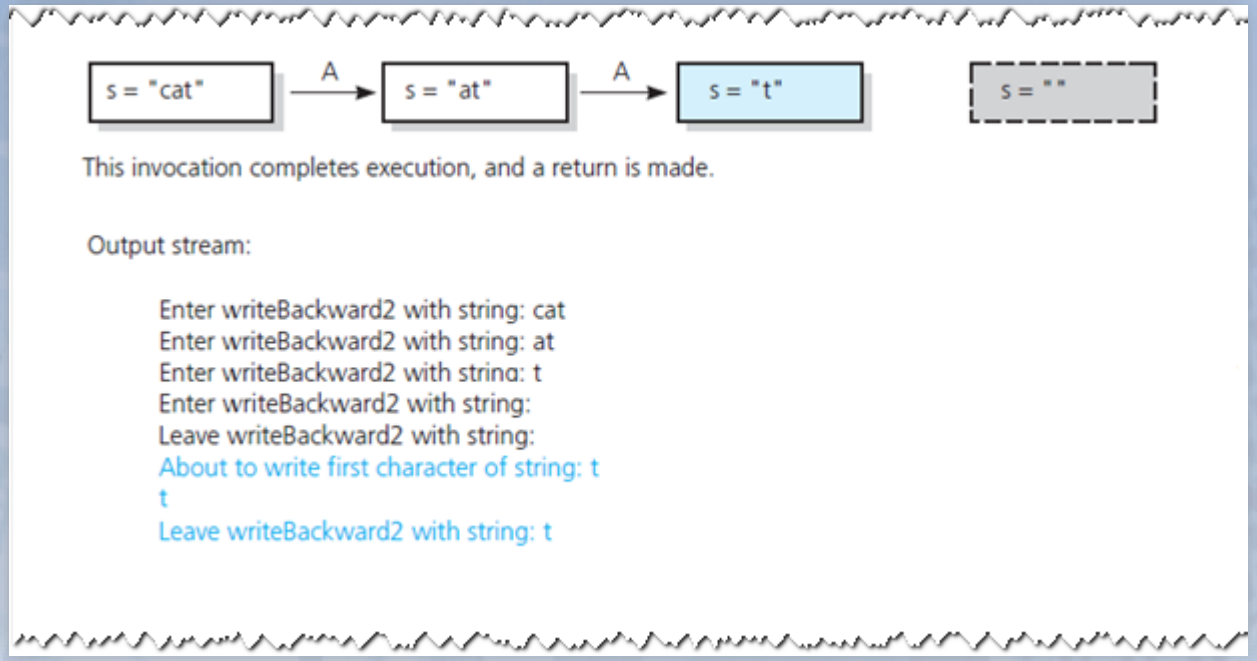

FIGURE 2-8 Box trace of writeBackward2("cat") in pseudocode

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A Recursive Void Function: Writing a String Backward

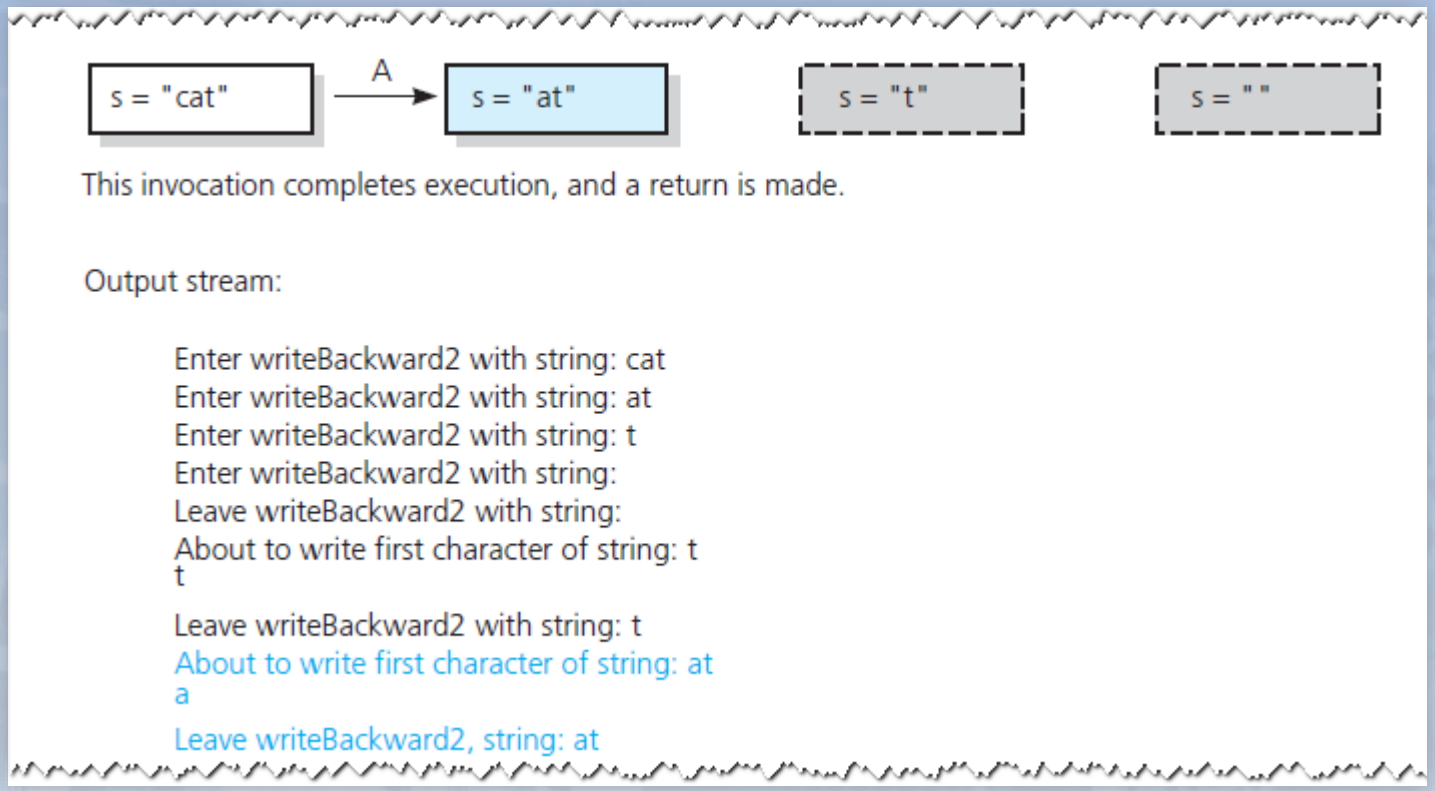

FIGURE 2-8 Box trace of writeBackward2("cat") in pseudocode

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A Recursive Void Function: Writing a String Backward

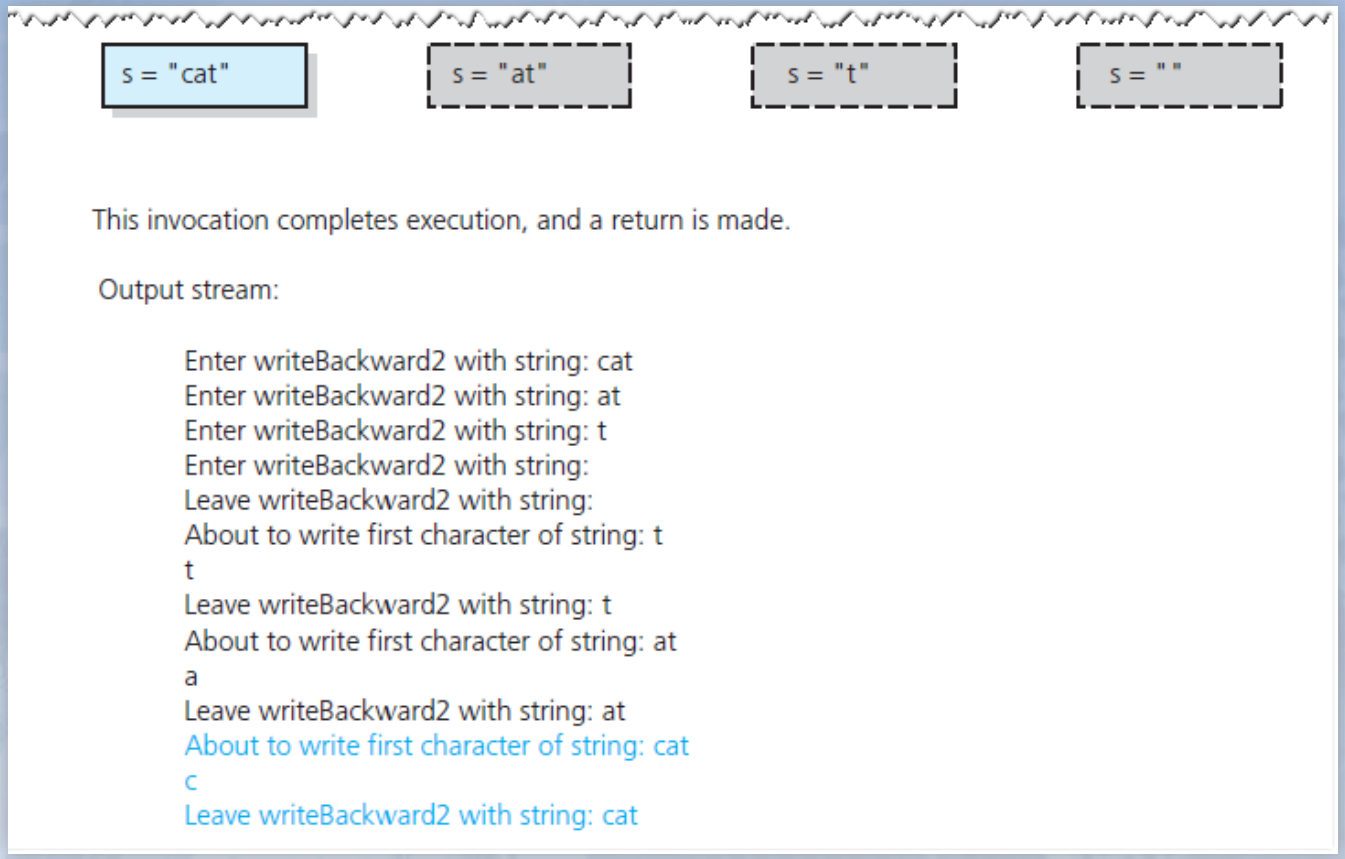

FIGURE 2-8 Box trace of writeBackward2("cat") in pseudocode

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Writing an Array's Entries in Backward Order

```
/** Writes the characters in an array backward.
  @pre  The array anArray contains size characters, where size >= 0.
  @post  None.
  @param anArray  The array to write backward.
  @param first  The index of the first character in the array.
  @param last  The index of the last character in the array. */
void writeArrayBackward(const char anArray[], int first, int last)
{
   if (first <= last)
   {
      // Write the last character
      cout << anArray[last];

      // Write the rest of the array backward
      writeArrayBackward(anArray, first, last - 1);
   }  // end if

   // first > last is the base case - do nothing
}  // end writeArrayBackward
```

The function writeArrayBackward

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The Binary Search

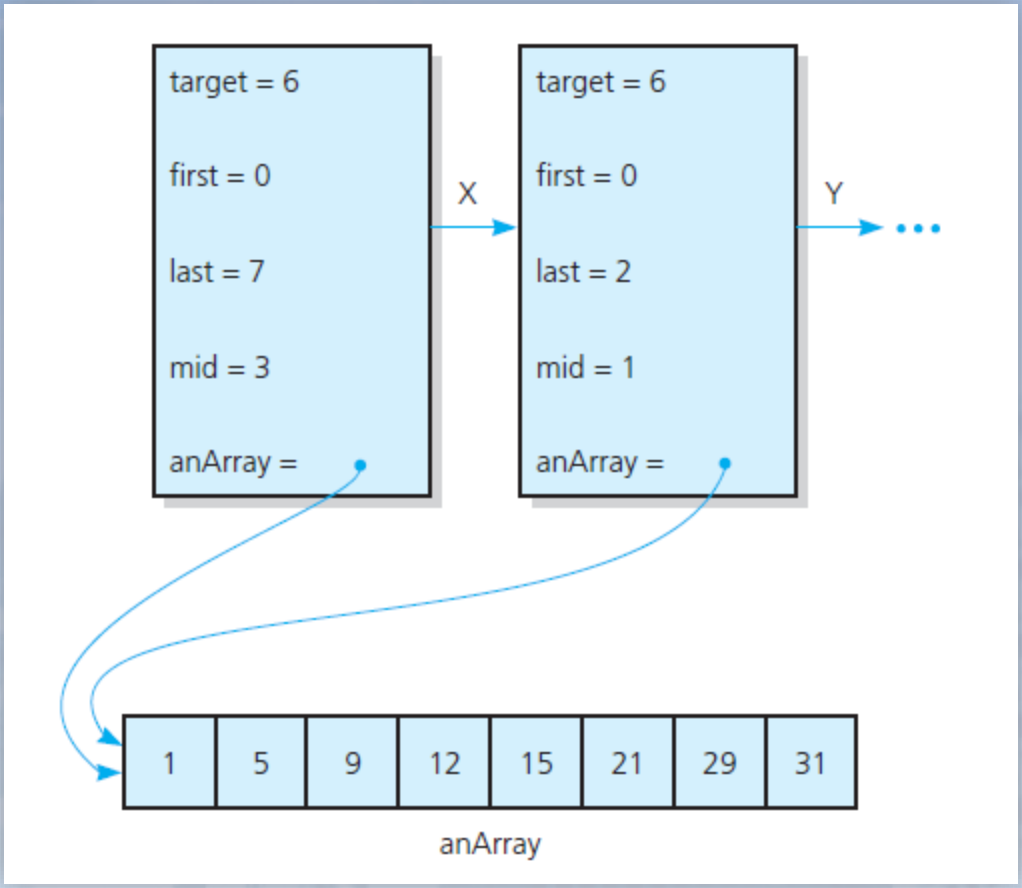

FIGURE 2-11 Box trace with a reference argument

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Finding the Largest Value in an Array

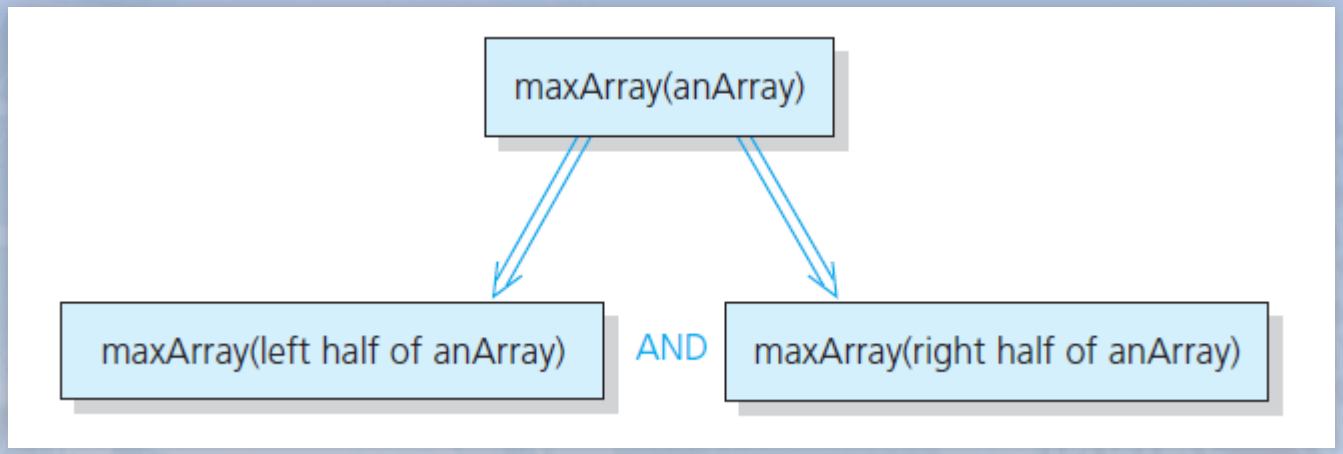

FIGURE 2-12 Recursive solution to the largest-value problem

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Finding the Largest Value in an Array

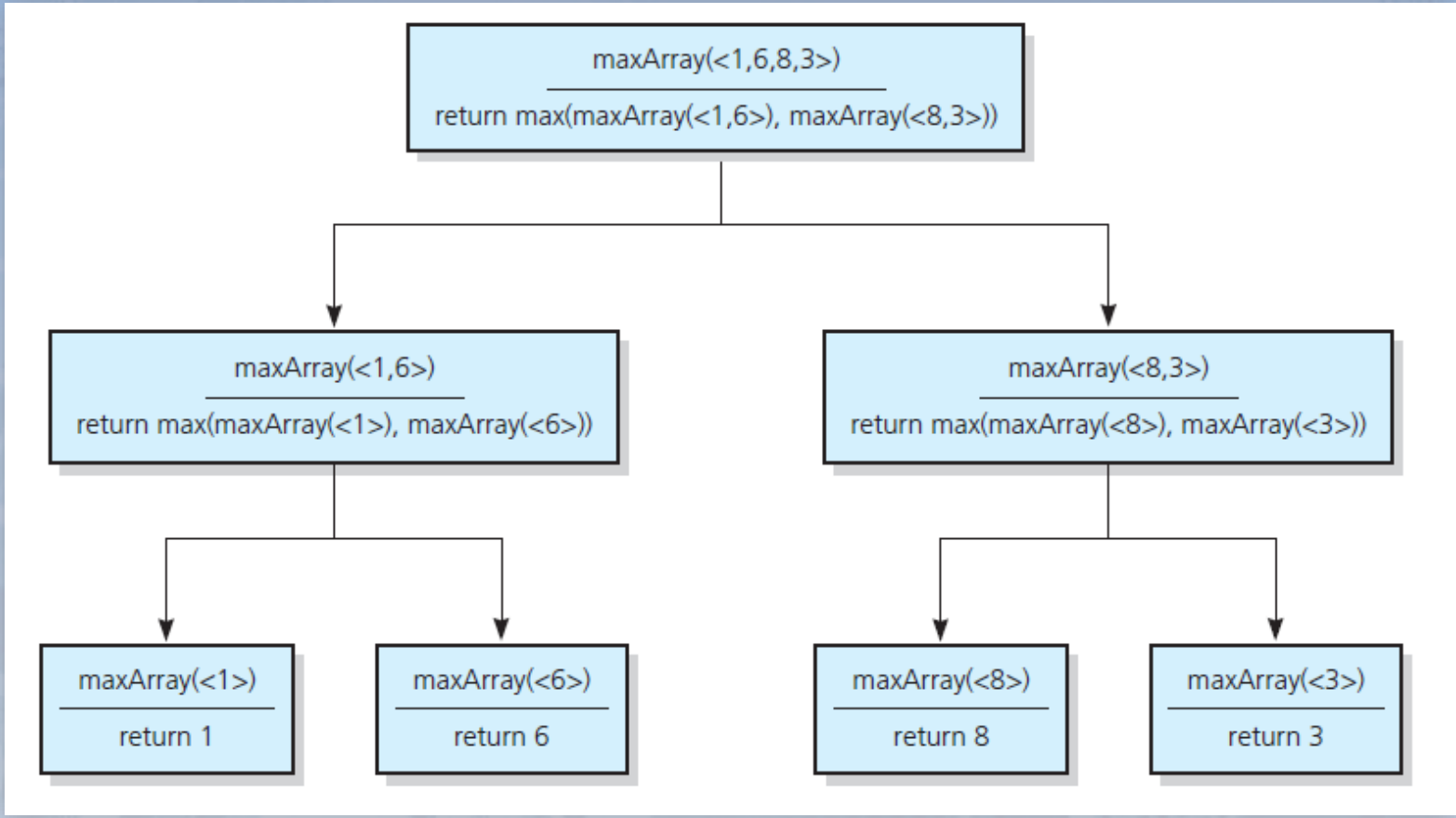

FIGURE 2-13 The recursive calls that maxArray(<1,6,8,3>) generates

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Finding k^{th} Smallest Value of Array

Recursive solution proceeds by:

1. Selecting pivot value in array
2. Cleverly arranging/ partitioning values in array about pivot value
3. Recursively applying strategy to one of partitions

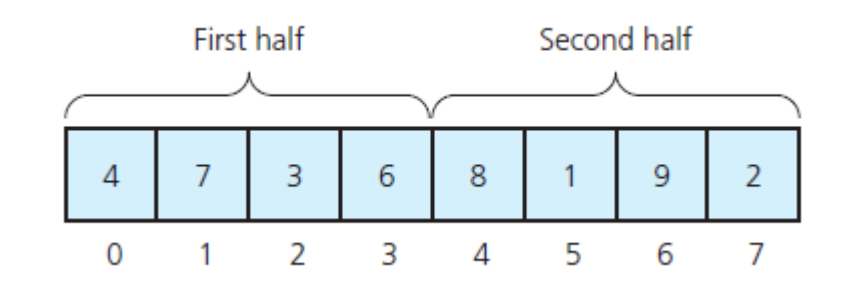

FIGURE 2-14 A sample array

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Finding k^{th} Smallest Value of Array

FIGURE 2-15 A partition about a pivot

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The Towers of Hanoi

- The problem statement
  - Beginning with *n* disks on pole A and zero disks on poles B and C, solve towers(n, A, B, C) .
- Solution
  1. With all disks on A, solve towers(n – 1, A, C, B)
  2. With the largest disk on pole A and all others on pole C, solve towers(n – 1, A, B, C)
  3. With the largest disk on pole B and all the other disks on pole C, solve towers(n – 1, C, B, A)

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The Towers of Hanoi

FIGURE 2-16 (

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The Towers of Hanoi

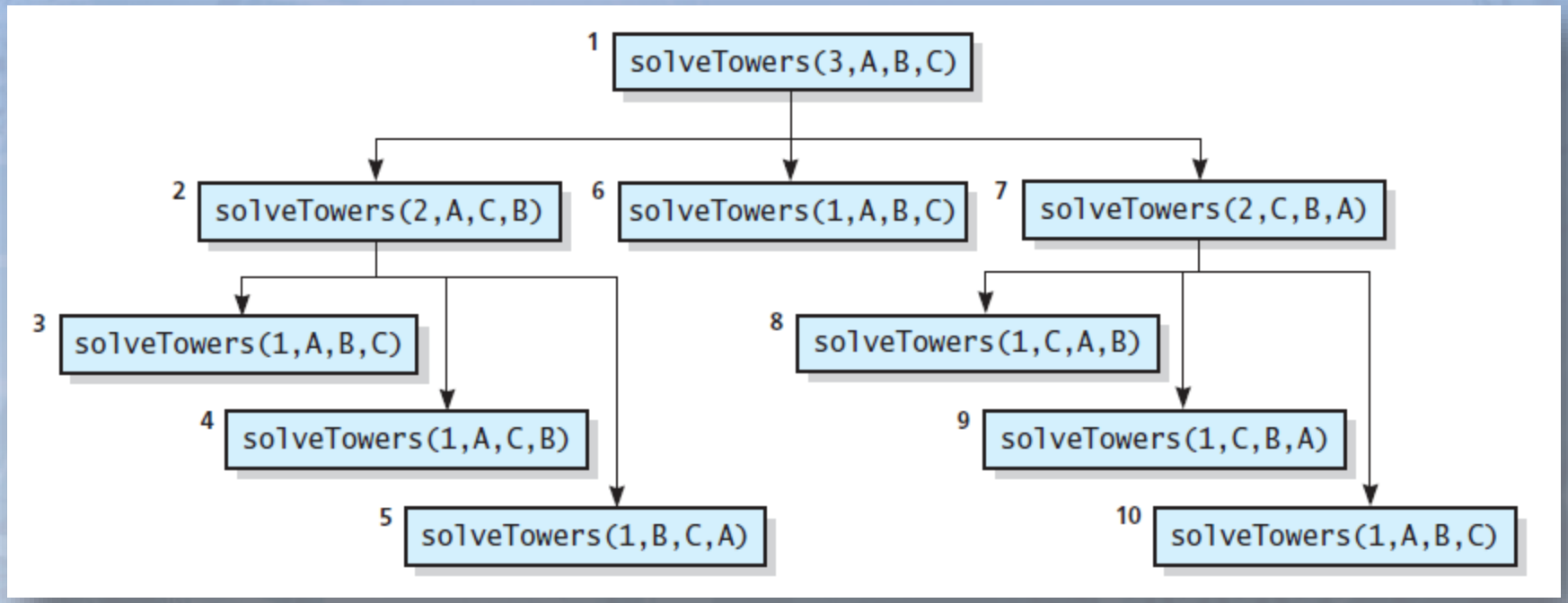

FIGURE 2-17 The order of recursive calls that results from solveTowers(3, A, B, C)

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The Fibonacci Sequence (Multiplying Rabbits)

Assume the following "facts" …

- Rabbits never die.
- Rabbit reaches sexual maturity at beginning of third month of life.
- Rabbits always born in male-female pairs. At beginning of every month, each sexually mature male-female pair gives birth to exactly one male-female pair.

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The Fibonacci Sequence (Multiplying Rabbits)

Monthly sequence

1. One pair, original two rabbits
2. One pair still
3. Two pairs (original pair, two newborns)
4. Three pairs (original pair, 1 month old, newborns)
5. Five pairs …
6. Eight pairs …

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The Fibonacci Sequence (Multiplying Rabbits)

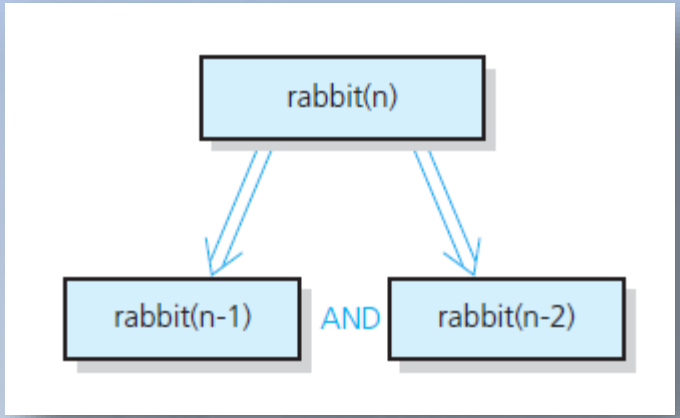

FIGURE 2-18 Recursive solution to the rabbit problem (number of pairs at month *n*)

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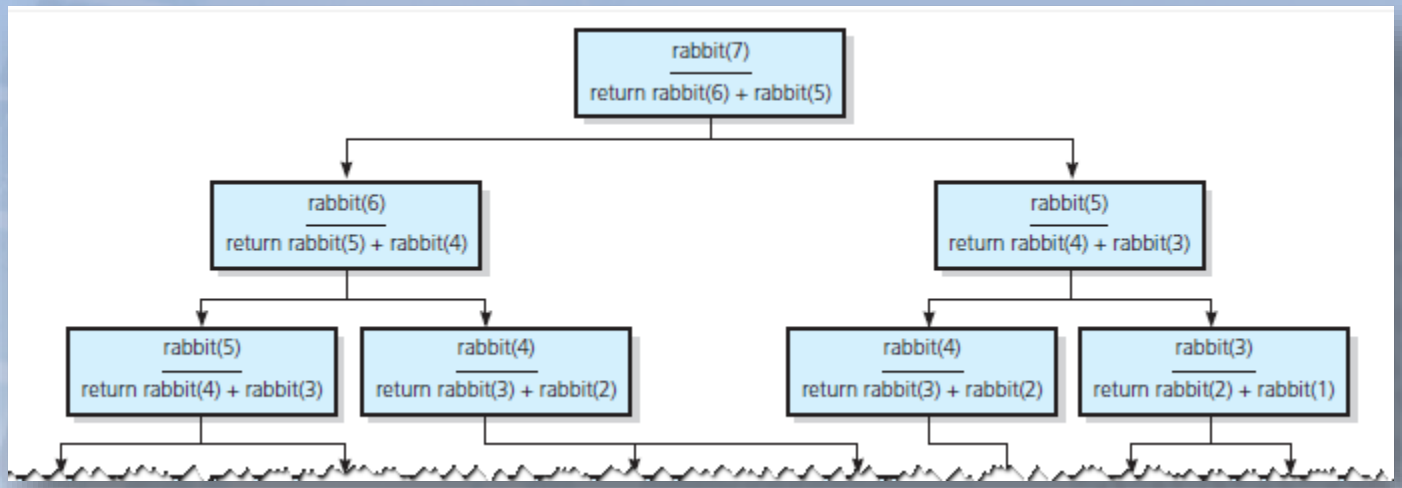
The Fibonacci Sequence (Multiplying Rabbits)



F IGURE 2-19 The recursive calls that rabbit(7) generates

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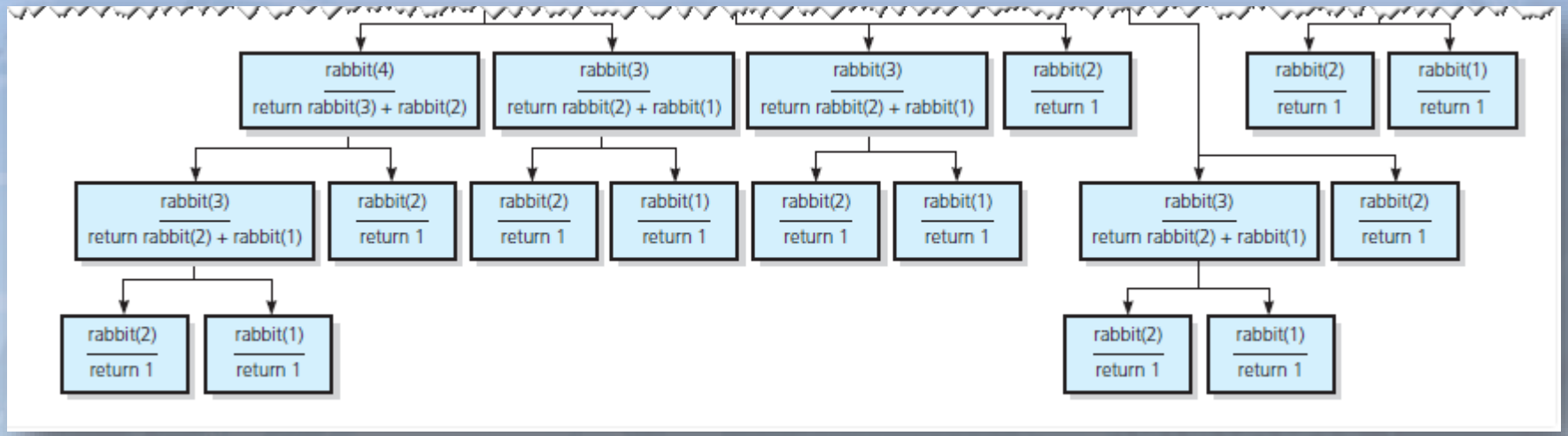
The Fibonacci Sequence (Multiplying Rabbits)



F IGURE 2-19 The recursive calls that rabbit(7) generates

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Organizing a Parade

- Will consist of bands and floats in single line.
  - You are asked not to place one band immediately after another
- In how many ways can you organize a parade of length n ?
  - $P(n)$ = number of ways to organize parade of length n
  - $F(n)$ = number of parades of length n, end with a float
  - $B(n)$ = number of parades of length n, end with a band
- Then $P(n) = F(n) + B(n)$

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Organizing a Parade

- Possible to see
  - $P(1) = 2$
  - $P(2) = 3$
  - $P(n) = P(n - 1) + P(n - 2)$ for $n > 2$
- Thus a recursive solution
  - Solve the problem by breaking up into cases

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Choosing *k* Out of *n* Things

- Rock band wants to tour *k* out of *n* cities
  - Order not an issue
- Let *g(n, k)* be number of groups of *k* cities chosen from *n*

$$g(n,k) = g(n-1,k-1) + g(n-1,k)$$

- *Base cases*

$$g(k,k) = 1$$
$$g(n,0) = 1$$

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Choosing *k* Out of *n* Things

```
/** Computes the number of groups of k out of n things.
 @pre  n and k are nonnegative integers.
 @post  None.
 @param n  The given number of things.
 @param k  The given number to choose.
 @return  g(n, k). */
int getNumberOfGroups(int n, int k)
{
   if ( (k == 0) || (k == n) )
      return 1;
   else if (k > n)
      return  0;
   else
      return getNumberOfGroups(n - 1, k - 1) + getNumberOfGroups(n - 1, k);
}  // end getNumberOfGroups
```

Function for recursive solution.

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Choosing *k* Out of *n* Things

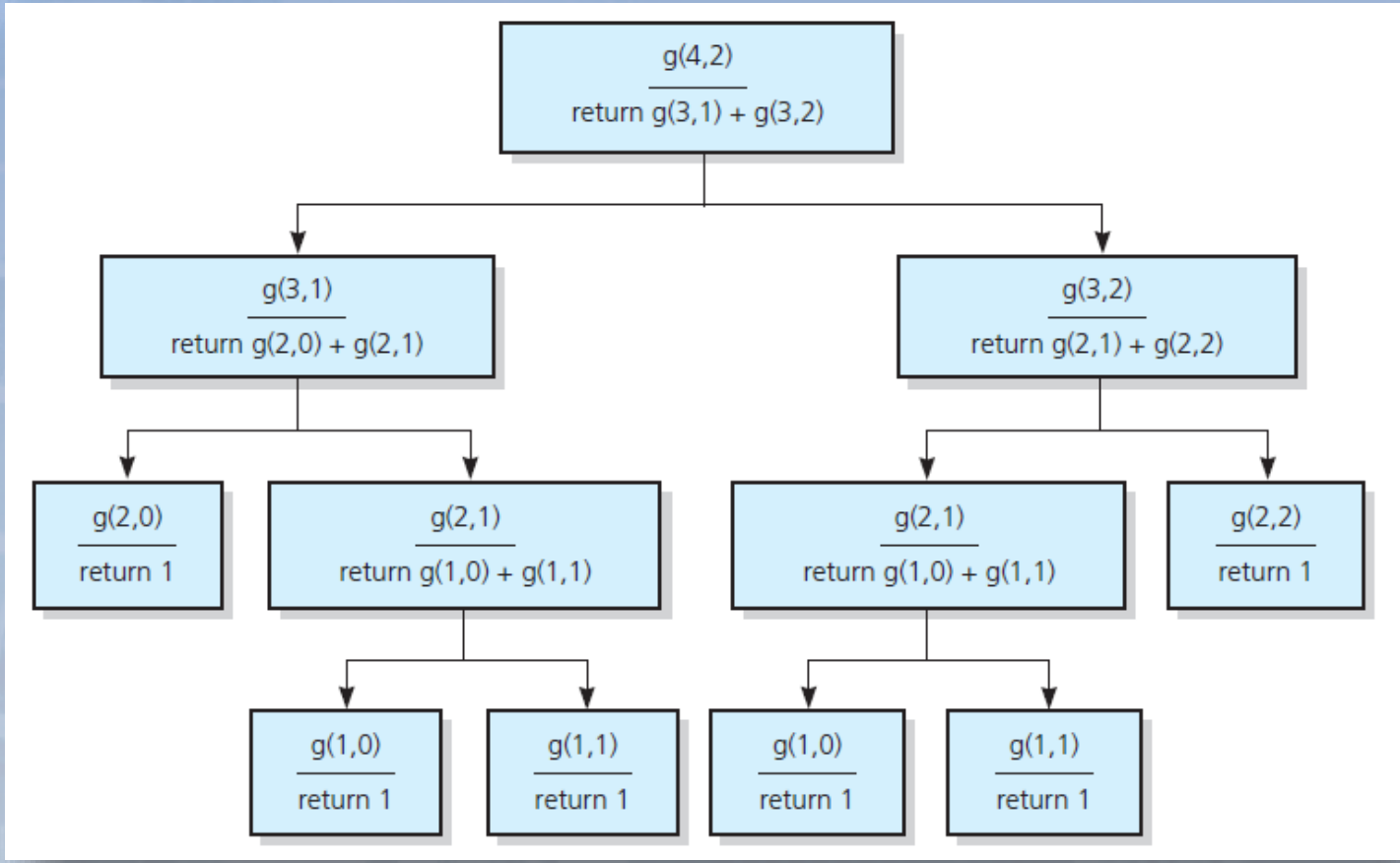

FIGURE 2-20 The recursive calls that g (4, 2) generates

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